Physical-Gram Coulomb Compression and Continuum Certificates

A local refinement and dependency audit of M557

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Technical Report Assisted by AI Models
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Technical Report TR-Ram-OpenAI-722-M557

October 7, 2026 (audit time: October 8 UTC)

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Abstract

We audit the pinned manuscript *Continuum Coulomb hardness with binary nuclear charges*, distinguishing its finite spin reductions, one-well computation, relative tunneling calibration, physical orbital metric, and exclusion of all omitted continuum states. We prove an isometric two-particle Coulomb comparison which, using the manuscript's stated orbital and separation estimates, replaces its coefficient-counting compression error Cm^5/D by $Cm^{7/2}/D$. The proof normalizes disjoint hydrogenic cutoffs, controls the physical H^1 synthesis map, applies Hardy's inequality before particle-number lifting, and retains intersite density repulsion. A positive angular quadrature reduces the sufficient spatial sample count from $O((D/\epsilon)^4)$ to $O((D/\epsilon)^3)$ with certified rational weights; this is not a total-runtime claim. We also prove a sharp scalar two-block lower bound and a physical-Gram residual-matrix certificate for semibounded continuum forms. These local improvements neither independently establish the source hardness theorem nor provide a polynomial-time many-electron solver; the source's uniform well, calibration, and bit-complexity inputs remain separately identified dependencies. The relevant private Ram passages concern metric transport, full residuals, and checked elimination, while the analytic methods are classical. Contribution assessment: C3/5, proved improvements in Coulomb-compression dimension loss and positive quadrature node count with explicit complement certificates give notable local gains while the global hardness construction and its analytic inputs remain unverified independently.

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1 Pinned problem and exact scope

1.1 Source identity and reading boundary

The assigned catalogue row is ID 557, with catalogue/source pin `adc7f1241b42e322a645 1854ab7e4b4c146bf78a`. The source is OpenAI's September 24, 2026 manuscript *Continuum Coulomb hardness with binary nuclear charges* [M557]. The local source root is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/
Continuum-Coulomb-hardness-with-binary-nuclear-charges-September-24-2026/
```

We read the main file, preamble, bibliography, and every substantive section and proof in `build/sections/00-introduction.tex` through `build/sections/05-completion.tex`. There are no separately included mathematical appendices in this source: the analytic, numerical, arithmetic, and continuum proofs are in those sections. This is a full-source reading, not an independent proof of every assertion. The local snapshot has no Git metadata, so the pin is recorded from the catalogue; file-level SHA-256 hashes identify the actual local inputs rather than pretending that a local Git revision was verified.

The report separates three levels of evidence: identities and local theorems proved here; source assertions used as explicitly named hypotheses; and proposed computations not executed. Section 10 and the companion audit record specify executed finite checks. No molecular continuum instance was numerically solved, and no performance comparison was measured.

1.2 Hilbert space, domains, and the promise

For $N \geq 1$ let

$$\mathfrak{h} = L^2(\mathbb{R}^3;\mathbb{C}^2), \qquad \mathfrak{H}_N = \bigwedge^N \mathfrak{h}, \qquad \mathcal{Q}_N = H^1(\mathbb{R}^{3N};(\mathbb{C}^2)^{\otimes N}) \cap \mathfrak{H}_N.$$

Inner products are conjugate linear in the first argument. Antisymmetry permutes spatial and spin variables together. Norms and orthogonality are the physical Lebesgue L^2 ones, including the spin sum. All spin sectors are admissible. The clamped-nucleus electronic operator in atomic units is

$$H = -\frac{1}{2}\sum_{\ell=1}^{N} \Delta_{x_\ell} - \sum_{\ell=1}^{N}\sum_{\alpha=1}^{M} \frac{Z_\alpha}{|x_\ell - R_\alpha|} + \sum_{\ell<k} \frac{1}{|x_\ell - x_k|}. \tag{1.1}$$

For each finite instance, translated Hardy inequalities and Fourier interpolation make the Coulomb multipliers infinitesimally operator bounded relative to the Laplacian. Thus the self-adjoint realization has operator domain $H^2 \cap \mathfrak{H}_N$ and closed form domain $\mathcal{Q}_N$. Equivalently one can specify it directly by its lower-bounded closed form. No finite box, Dirichlet boundary, periodic boundary, ultraviolet cutoff, or condition at a finite nuclear surface is imposed. Behavior at infinity is governed by membership in these spaces, not an assumed binding eigenfunction. Define

$$E_0 = \inf \operatorname{spec} H = \inf\{q_H[u] : u \in \mathcal{Q}_N,\ \|u\| = 1\}.$$

This variational infimum need not be attained.

The inputs are $M \geq 1$ pairwise distinct rational positions $R_\alpha \in \mathbb{Q}^3$, positive integer charges Z_α encoded in binary, unary N, and rational $a < b$. Rational numerators and positive denominators are in lowest terms with the source's fixed self-delimiting binary encoding; L denotes total input length. For the main theorem the threshold promise is $b - a \geq L^{-1}$, YES means $E_0 \leq a$, and NO

means $E_0 \geq b$. The source claims a deterministic classical polynomial-time many-one reduction from every QMA promise problem, producing $b - a \geq 1$. This is a threshold separation, not a molecular spectral gap. There is no neutrality, fixed-element, bounded-charge, minimum-separation, or binding promise. Nuclear repulsion is omitted by the electronic-energy convention, and no orbital basis, magnetic field, or extra adjustable potential is an input. Membership in QMA is not claimed.

1.3 Three energy units and two different projections

The source constructs m sites, two nuclei per site, and $N = m$ electrons. Put $Z_* \asymp e^D$, where D is a fixed polynomial in source size. The unitary dilation from physical to scaled coordinates is

$$(\mathcal{D}_{Z_*} u)(y_1, \ldots, y_m) = Z_*^{-3m/2} u(y_1/Z_*, \ldots, y_m/Z_*),$$

with spins unchanged. In the following we identify the operators by this unitary. If h is the scaled one-body operator, $A = h + 1/2$, and $W = \sum_{\ell<k} |y_\ell - y_k|^{-1}$, then

$$\mathcal{H} := H/Z_*^2 + m/2 = d\Gamma(A) + W/Z_*, \qquad Z_* \inf \operatorname{spec} \mathcal{H} = E_0/Z_* + mZ_*/2. \tag{1.2}$$

Here $d\Gamma(B) = \sum_{\ell=1}^m B^{(\ell)}$ on fixed particle number. The finite low-orbital space has dimension $\binom{2m}{m}$, not m. Its projection Π retains all configurations, including double occupancy. The further singly occupied projection P_{so} is used only in the Hubbard perturbation argument. Confusing Π and P_{so} would change the continuum problem.

2 Stepwise source dependency audit

The file abbreviations S0–S5 mean the six numbered files under `build/sections/`. Literal source labels below are locating keys, not labels of this report. Each stage is read in its full local proof context. The audit records what a downstream argument needs, rather than certifying the whole stage by a restatement.

2.1 S1: signed spins, positive mediators, and Hubbard exchange

The imported complexity input, `thm:source`, is a signed rational Heisenberg problem on finite nonwrapping square-lattice subgraphs. The Pauli convention is

$$h_{ij}^s = X_i X_j + Y_i Y_j + Z_i Z_j, \qquad \operatorname{spec} h_{ij}^s = \{-3, 1\}.$$

Cubitt–Montanaro–Piddock's Theorem 48 and spatially sparse branch of Lemma 47 are the relevant public inputs [CMP]; the complete external complexity proof is not reproduced here. The source's rounding step uses $\|h^s\| = 3$, shifts the thresholds for coefficient and scalar errors, and preserves an inverse-polynomial gap δ_s. It does not assume a lower bound on each nonzero signed edge weight.

The finite second-order lemma `lem:second-order` assumes $A_f \geq gQ_f$, $P_f = \ker A_f \neq 0$, $P_f B_f P_f = 0$, and $\|B_f\| \leq \epsilon g$ with $\epsilon < 1/2$. It bounds the bottom-energy difference from $-P_f B_f A_f^{-1} B_f P_f$ by $2g\epsilon^3$. The inverse vanishes on P_f. The proof's lower bound completes a square with $(1-\epsilon)A_f$; its trial vector for the upper bound includes its norm correction. The empty-complement case is exact.

For each source edge $e = LR$, $v_e = \sqrt{2|J_e|}$ and an ancillary pair has singlet–triplet gap 4Δ. Coupling both endpoints to the same ancilla gives negative exchange; coupling opposite ancillas gives positive exchange. The second-order contribution is

$$J_e h_{LR}^s - \tfrac{3}{2} v_e^2 I.$$

Thus `prop:positive-spin` has error $27(\sum_e v_e)^3/\sqrt{\Delta}$ and scalar $C_0 = 3k\Delta + \frac{3}{2}\sum_e v_e^2$; the scalar cannot be discarded in the final thresholds.

The geometric contact lemma includes an extra unit contact between consecutive positive gadgets. Those contacts receive zero spin weight but a small positive Hubbard hopping floor. All other distinct noncontacts are separated by at least $\sqrt{2}$. The constructive right inverse of linearized contact length has sup-norm bound $6m$, using independent row and column scalar paths; it does not solve an exact nonlinear prescribed-length system. Rational rounding leaves the exact vector identity $X_j - X_i = Du_{ij} + s_j - s_i$, and the length expansion has a separately charged $O(S^2/D)$ error.

In the half-filled Hubbard space the onsite constant is $U = 5/8$, and the two-hop calculation gives

$$-P_{\mathrm{so}} T_{ij} A_U^{-1} T_{ij} P_{\mathrm{so}} = \frac{t_{ij}^2}{U}(h_{ij}^s - I), \qquad A_U = U \sum_i n_{i\uparrow} n_{i\downarrow}. \tag{2.1}$$

The factor uses Pauli matrices, not spin operators divided by two. Different bonds have zero compressed two-hop cross terms. The source chooses

$$\bar{R} = \max(1, |\mathcal{C}|\Delta), \quad \eta = \frac{U\gamma}{1024\bar{R}^3}, \quad \alpha = \eta^2/U, \quad f = \frac{\alpha\gamma}{128\max(1,|\mathcal{C}|)}, \quad \gamma = \min(1, \delta_s).$$

Rational $t_{ij} = \max(f, r_{ij})$, with $|r_{ij} - \eta\sqrt{K_{ij}}| \le f$, satisfy

$$|E_{\mathrm{Hub}} - \alpha(E_{\mathrm{in}} - C_s)| \le 3\alpha\gamma/64, \qquad C_s = C_0 + \sum_{\mathcal{C}} K_{ij}. \tag{2.2}$$

The finitely many sign and normalization identities in this stage can be tested exactly. Such tests do not establish the imported QMA-hardness theorem.

2.2 S2: well isolation, graded approximation, and certified tuning

The scaled charges are fixed by integer rounding before any tuning:

$$Z_* q = \mathrm{round}(Z_*(1 - 500m/D)), \qquad Z_* q' = \mathrm{round}(Z_* 1000m/D).$$

With $\rho_i \in [1,4]$, the own potential is $V_i = q/r_i + q'/|y - X_i - \rho_i e_z|$. The globally C^2 caps $\chi_i = \chi(r_i/D)$ vanish for $r_i \le D/16$ and equal one for $r_i \ge D/8$. The isolated operator

$$h_i = -\Delta/2 - V_i - W_i, \qquad W_i = \sum_{k \ne i} \chi_k V_k$$

retains remote fields near its own well and removes remote singularities. These caps are auxiliary, never physical output potentials.

`prop:isolated-well` asserts a simple positive normalized real ground mode $\psi_i \in H^2$, a uniform gap $g_{\mathrm{w}} > 0$, and

$$\|\psi_i - \phi_i\|_{H^2} \le Cm/D, \quad \phi_i = \pi^{-1/2} e^{-r_i}, \quad \left\| D^j W_i \right\|_\infty \le CmD^{-j-1} \quad (0 \le j \le 2).$$

Its proof combines Hardy, graph-norm control, compactness for negative-energy attainment, and the hydrogenic square identity. These uniform analytic constants are inherited inputs to our source specialization; the abstract local theorems below do not depend on their derivation.

`lem:wells-secants` controls the moving secondary pole through expectation-level Hardy estimates, not an incorrectly bounded derivative multiplier. The own-coordinate slope is between cq'

and Cq', while the other-coordinate Lipschitz constant is $Cq'm/D^2$. Endpoint values bracket $-1/2$. Tail estimates give pointwise $Ce^{-.99(r_i-.01D)}$ and exterior H^1 norm at most $e^{-.61D}$ beyond $.65D$.

The analytic-patch lemma bounds factorial derivatives on pole-free rescaled balls. The graded rational tensor grid uses $\delta = 2^{-8D}$ and degree $L_0 = O(D)$, with $O(D^6)$ candidate basis functions formed from hats times monomials. Node deletion around poles and an exterior cutoff give an H^1 approximant error $e^{-.58D}$, then a one-sided Ritz bias $Ce^{-1.16D}$ by eigen-equation cancellation. The proof explicitly treats skinny cells by coordinate secants; no shape-regular mesh theorem is silently substituted.

The oracle `prop:ritz-oracle` removes redundant basis vectors by exact rational mass-matrix elimination. If r is the reduced dimension and B_0 an entry bit bound, its conservative certificate is

$$M_{\mathrm{Ritz}} \succeq 2^{-K_M} I, \qquad K_M = r^3 B_0 + (r-1)(B_0 + \lceil \log_2 r \rceil).$$

Potential entries use convergent pole-free binomial expansions on rational boxes. Entry tolerance $\varepsilon_D 2^{-K_M}/(16r)$ becomes a metric operator error at most $\varepsilon_D/16$, where $\varepsilon_D = 2^{-\lceil 8D/5 \rceil}$. Exact inertia bisection on $\widetilde{A} - zM_{\mathrm{Ritz}}$ avoids an eigenvalue-separation assumption. The lower enclosure endpoint subtracts the Ritz bias; a Ritz upper bound alone is insufficient. Polynomial dimension without this arithmetic analysis would not certify polynomial bit cost.

Finally the simultaneous root map has contraction factor $Cm/D^2 \le 1/2$. Certified interval bisection either decides a sign or stops when the energy interval contains $-1/2$; $O(mD^2)$ oracle calls produce rational ρ_i with $O(D)$ fractional bits and

$$|\lambda_i + 1/2| \le e^{-1.01D}. \tag{2.3}$$

These are source claims with quantitative proofs read, not computations executed in this report. Our refinement does not weaken the tuning accuracy requirement.

2.3 S3: relative tails and noncircular hopping calibration

An absolute $O(m/D)$ orbital error cannot estimate an e^{-D} overlap relatively. The ray primitive

$$F_i(y) = r_i \int_0^1 W_i(X_i + t(y - X_i))\,\mathrm{d}t$$

satisfies $\partial_{r_i} F_i = W_i$. In the tuned setting the source proves, uniformly for $r_i \le 2D$,

$$\psi_i(y) = \pi^{-1/2} e^{-r_i + F_i(y)} \left(1 + O\left(\frac{m \log D + m^2}{D}\right)\right).$$

The proof uses the residual quotient and barriers $e^{\pm g(r)}$ on $10 \le r \le 3D$, with an exponentially small additive outer-boundary correction. This is a relative estimate in a fixed region, not a global asymptotic eigenfunction expansion.

Before secondary positions are known, capped unit-primary surrogate fields W_i^* replace W_i with sup error Cm^2/D^2 and ray error Cm^2/D. The positive attraction surrogate $T_{ij}(X)$ is integrated with weight $(1 - \chi(|v|/D))/|v|$; that singular but integrable weight is not logarithmically differentiated. For reference contacts,

$$ce^{-D} \le T_{ij}(Dp) \le CD^2 e^{-D+Cm}.$$

Logarithmic oscillation estimates and exact positive radial/angular weights give relative quadrature for $e^D T_{ij}$ in time polynomial in D, m, ϵ^{-1}, not a claim of polynomial dependence on $\log \epsilon^{-1}$.

The displacement law retains the non-negligible backward segment in the integral and bounds transverse contributions. With $S = \max_i |s_i|$ and $m + S \le D^{1/4}$, its error is

$$\frac{mS}{D} + \frac{S^2}{D} + SD^{-1/4} + e^{-c_1\sqrt{D}}.$$

Calibration sets $u_{ij} \cdot (s_j - s_i)$ to an approximation of $\log(Z_* T_{ij}(Dp)/t_{ij})$. The identity $Z_* T_{ij}(Dp) e^{-u_{ij}\cdot(s_j - s_i)} = t_{ij} e^{-e_{ij}}$ with $|e_{ij}| \le \tau$ avoids an uncontrolled e^{Cm} error amplification. The exact orientation relation

$$B_{ij} - B_{ji} = (\lambda_i - \lambda_j)\langle \psi_j, \psi_i \rangle$$

costs at most $4e^{-.86D}$ after multiplying by Z_*. The algorithm fixes and rounds primary centers first, then tunes secondaries at those centers. No subsequent center or charge rounding is allowed without re-estimating the tuned energy.

2.4 S4: the physical metric and all-state lower bound

The synthesis map $\mathcal{U} e_i = \psi_i$ has Gram matrix $G = \mathcal{U}^*\mathcal{U}$ and exact isometric orthonormalization $\mathcal{J}_0 = \mathcal{U} G^{-1/2}$. The source's residual/Gram lemma gives

$$\|A\psi_i\| \le e^{-.75D}, \quad \|G - I\| \le e^{-.85D}, \quad \|AP_0\| \le C\sqrt{m} e^{-.75D}, \quad \|\omega_i - \phi_i\|_{H^1} \le Cm/D, \tag{2.4}$$

where $P_0 = \mathcal{J}_0 \mathcal{J}_0^*$ and $\omega_i = \mathcal{J}_0 e_i$. The raw shifted matrix is

$$M = \mathcal{U}^*(h + 1/2)\mathcal{U} = \mathcal{U}^* h \mathcal{U} + \tfrac{1}{2} G, \tag{2.5}$$

not $\mathcal{U}^* h \mathcal{U} + I/2$. With target hopping matrix T, diagonal zero and contact entries $-t_{ij}$,

$$\begin{aligned} \|Z_* \mathcal{J}_0^* A \mathcal{J}_0 - \mathsf{T}\| &\le \delta_1, \\ \delta_1 &= m\epsilon_{\mathrm{hop}} + Z_* e^{-1.01D} + CmZ_* e^{-1.1D} + CmZ_* e^{-1.6D}. \end{aligned} \tag{2.6}$$

The $e^{-1.6D}$ term is the product of Gram and shifted-residual errors; estimating the unshifted operator first loses this benefit.

For spin, $P = P_0 \otimes I_2$, $\mathcal{J} = \mathcal{J}_0 \otimes I_2$, and $\Pi = \bigwedge^m P$. The IMS proof of `prop:complement-gap` uses disjoint $D/4$ balls, equality $h = h_i$ on their supports, a weak exterior attraction Cm/D, and a kinetic localization error CD^{-2}. Global orthogonality gives local ground-mode leakage at most $me^{-.36D}$; local orthogonality is not assumed. It yields $(I - P)A(I - P) \ge g(I - P)$ for the whole complement, including infinity.

The kinetic trace, not the exponentially large configuration dimension, gives $\sum_\ell \|\nabla_\ell u\|^2 \le Cm$ for every normalized $u \in \operatorname{Ran}\Pi$. Hardy then gives $\|W\Pi\| \le Cm^2$. Put $\varepsilon = \|AP\|$ and $Q_m = I - \Pi$. Counting high particles and using $W \ge 0$ gives

$$Q_m \mathcal{H} Q_m \ge (g - 2m\varepsilon) Q_m, \qquad \|Q_m \mathcal{H} \Pi\| \le m\varepsilon + Cm^2/Z_*. \tag{2.7}$$

The source completes a square to bound the energy loss by $2\beta^2/c$, eventually at most $e^{-1.2D}$. Section 7 sharpens this step without changing its analytic prerequisites.

The source's `lem:continuum-coulomb-compression` replaces four-index coefficients individually at cost Cm/D each and sums $O(m^4)$ monomials. This gives Cm^5/D, followed by disjoint-cutoff onsite/direct-density estimates. Section 5 replaces this particular coefficient-counting stage, not the well or gap proofs.

2.5 S5: global scale, encoding, and the two promises

The source chooses $D = A_0(n+2)^K$ with fixed sufficiently large K, A_0 and inverse-polynomial τ. If $S \le Cn^b(1+\log D)$, all conversion losses are at most Cn^h, and $\epsilon_* \ge cn^{-r}$, its sufficient choice is $K > 4(b+h+r+2)$. The logarithmic dependence of S is handled after fixing K, so the choice is not an input-dependent search for a good scale. We do not compute numerical values of the universal constants.

The final molecular centers are X_i/Z_* and $(X_i+\rho_i e_z)/Z_*$, with $N = m$, positive integer charges, and $2m$ distinct nuclei. Charges can be exponentially large, but require $O(D+\log n)$ bits. A rational truncated exponential sum supplies a suitable integer Z_*; no exact transcendental comparison is needed.

The continuum budget is $\alpha\gamma/64$; together with (2.2) it gives $\alpha\gamma/16$ in Hubbard units. If $|\widetilde{C}_s - C_s| \le \delta_s/64$, define

$$\widetilde{F}(z) = -mZ_*^2/2 + Z_*\alpha(z-\widetilde{C}_s), \qquad (2.8)$$
$$a = \widetilde{F}(a_s+\delta_s/4), \qquad b = \widetilde{F}(b_s-\delta_s/4).$$

The total affine error is at most $5Z_*\alpha\delta_s/64$, below the 1/4-gap threshold allowance; $b-a = Z_*\alpha\delta_s/2 \ge 1$. This transfer requires the full-continuum lower bound. A finite variational energy alone would transfer only the YES direction. All these downstream consequences remain conditional on the source analytic and complexity interfaces.

3 Actual private Ram correspondences

The following passages were read in the original local files. Routing maps did not substitute for those readings. Their role is representational discipline and checked inference, not a new physical interaction or an easier many-electron decision problem.

3.1 Volume 21: metric, residual, domain, and exclusion

The common private directory is

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/`

In `ch01d-physical-metrics.tex`, the sections "A Nonorthogonal Representation Carries a Metric", "The Full Residual Has an Omitted Component", and "Approximating the Metric" give the exact correspondences

$$J \leftrightarrow \mathcal{U}, \quad J^*J \leftrightarrow G, \quad B = G^{-1}J^*AJ, \quad P = JG^{-1}J^*.$$

The locating labels are `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-identity`, and `v21c01:gram-equivalence`. Section 4 derives the source-specific residual matrix and scalar-shift consequences. The columnwise Sobolev estimate in Section 5 is also a physical synthesis-map estimate; it is not obtained by pretending that an arbitrary coefficient basis has Euclidean physical norm.

In `ch01a-state-domain.tex`, "Transporting a Closed Realization", label `v21c01:bounded-similarity`, requires a bounded bijection with bounded inverse and transportation of the operator domain. Here the dilation really is unitary on the continuum and transports its Sobolev domain. In contrast $\mathcal{U} : \mathbb{C}^m \to L^2$ is not surjective, so its Gram orthonormalization identifies only a finite range. It cannot transfer the whole continuum spectrum by similarity.

In `ch01f-spectral-limits.tex`, "A Dense Sequence With a Spurious Spectral Value", label `v21c01:pollution-proposition`, and "Strong Resolvents Do Not Give Uniform Spectral Exclusion", label `v21c01:strong-not-norm`, distinguish projected residuals from physical residuals. The counterexample is an indefinite diagonal model; it is not evidence that M557's semibounded variational minimum itself suffers that particular pollution. The directly relevant positive passage is "What a Stronger Contract Can Establish", label `v21c01:stronger-contract`, especially "Domain-aware approximate inversion": finite-rank inversion does not establish an inverse on an infinite space. M557 supplies a separate complement argument; our certificates explicitly retain it.

3.2 Core report 5: Schur reconstruction and limited reuse

The original read is

> `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`

The passage "Elimination is not pooling" and Proposition `prop:schurresidual`, equation `eq:schurresidual`, preserve the interior source and transport interior residual into retained rows. For a spectral trial energy z below a complementary lower bound, the analogous interior equation is

$$(H_Q - z)v = -Fc,$$

with the eliminated correction $-F^*(H_Q - z)^{-1}F$ in the finite form. Section 7 gives a complete form-level sufficient certificate without assuming a computable continuum inverse.

The theorem `thm:woodbury` in "Topology Edits and Checked Low-Rank Reuse" is explicitly classical low-rank algebra with conditioning and full-residual checks. Moving a Coulomb pole changes a multiplication operator of infinite rank in the continuum; it is not automatically a low-rank edit. A finite matrix update could be reused only after checking its actual rank structure and propagated errors. No Woodbury speedup is claimed here.

The passage "Polynomial evaluation without eigendecomposition", equation `eq:filtererror`, requires a self-adjoint finite matrix and a certified spectral interval. It could approximate a function of a whitened finite matrix. It cannot by itself furnish the missing continuum complement or replace the one-well oracle's mass lower bound. No polynomial filter was implemented in this report.

3.3 Attribution boundary

Physical Gram orthonormalization, Hardy inequalities, IMS localization, Rayleigh–Ritz, Schur complements, and Feshbach reduction are established mathematics. The private corpus organizes these around explicit state spaces, coordinate metrics, and full residuals. Our realized addition is their detailed application to M557's quantitative interface, including the two-particle compression proof. Neither use of a private Ram representation nor a better bound licenses a claim of historical priority. The source already uses the correct G shift and a full complement proof; this report does not present those existing strengths as newly repaired source errors.

4 Physical-Gram and residual certificates

Proposition 4.1 (Exact metric and omitted-residual matrix)**.** *Let T be self-adjoint in a Hilbert space $\mathcal{H}_0$, and let $S : \mathbb{C}^r \to \mathcal{H}_0$ be injective with columns in $D(T)$. Define*

$$G = S^*S, \quad K = S^*TS, \quad R = TS, \quad P = SG^{-1}S^*, \quad F = (I - P)R.$$

Then P is the orthogonal projection onto $\operatorname{Ran} S$*, $K = K^*$, and*

$$F^*F = R^*R - KG^{-1}K, \tag{4.1}$$

$$\|(T - z)Sc\|^2 = \left\|(G^{-1}K - z)c\right\|_G^2 + c^*F^*Fc \quad (z \in \mathbb{C}). \tag{4.2}$$

For $J = SG^{-1/2}$,

$$\|(I - P)TJ\|^2 = \left\|G^{-1/2}F^*FG^{-1/2}\right\|. \tag{4.3}$$

All statements are invariant under an invertible finite coordinate change $S \mapsto SC$ with the induced Gram congruence.

Proof. $G > 0$, and multiplication shows $P = P^* = P^2$ and $PS = S$. The symmetry of T gives $K = K^*$. Since $S^*R = K$, $R^*PR = KG^{-1}K$; subtracting from R^*R proves (4.1). The two orthogonal components of $(T - z)Sc$ are $S(G^{-1}K - z)c$ and Fc. Pythagoras gives (4.2). Finally the norm squared of a bounded map is the norm of its Gram matrix, proving (4.3). Under $S \mapsto SC$, each Gram/form/residual matrix changes by $C^*(\cdot)C$, and the physical projection is unchanged. Thus the residuals and physical induced norms are unchanged; the orthonormalized coordinates may differ by a unitary. □

4.1 Why the physical shift matters exponentially

Apply Proposition 4.1 with $T = A = h + 1/2$ and $S = \mathcal{U}$. The correct whitened matrix is $G^{-1/2}(\mathcal{U}^*h\mathcal{U} + G/2)G^{-1/2}$. If $I/2$ is inserted instead, the error is exactly

$$\tfrac{1}{2}(G^{-1} - I). \tag{4.4}$$

For $\rho = \|G - I\| < 1$, $\left\|G^{-1} - I\right\| \le \rho/(1 - \rho)$. Even if $\rho = e^{-.85D}$, multiplying this bound by $Z_* \asymp e^D$ permits exponential growth $e^{.15D}$. In contrast, if $M = \mathcal{U}^*A\mathcal{U}$ is small, write $L = G^{-1/2} - I$ to obtain

$$\left\|G^{-1/2}MG^{-1/2} - M\right\| \le (2\|L\| + \|L\|^2)\|M\|. \tag{4.5}$$

The source has $\|L\| = O(e^{-.85D})$ and $\|M\| = O(\sqrt{m}e^{-.75D})$, hence the $e^{-1.6D}$ margin. A diagnostic exact example is $h = -I/2$ on a nonorthogonal trial range: the true shifted matrix is zero, while the wrong prescription produces (4.4). This detects a coordinate error, not a source error.

4.2 Validated matrix errors require a mass margin

If a Hermitian computed matrix $\widehat{K}$ satisfies $\left\|\widehat{K} - K\right\| \le \eta$ and $G \succeq \mu I > 0$, then

$$\left\|G^{-1/2}(\widehat{K} - K)G^{-1/2}\right\| \le \eta/\mu.$$

Proof follows from submultiplicativity and $\left\|G^{-1/2}\right\| \le \mu^{-1/2}$. This is precisely the logical role of the source Ritz mass certificate, not an optional numerical convenience. If G is also approximated, a separate Loewner enclosure such as $(1 - \theta)G \preceq \widehat{G} \preceq (1 + \theta)G$, $\theta < 1$, is needed to transfer physical norms. Entrywise accuracy without the small-eigenvalue margin is insufficient.

The difference $R^*R - KG^{-1}K$ can suffer cancellation numerically. For a rigorous leakage upper bound one must enclose the full expression, or enclose F directly, and then bound its whitened positive form. Observing tiny floating-point eigenvalues of a subtracted matrix is not a certificate. No such validated Coulomb residual matrix has been computed here.

5 Isometric two-particle Coulomb comparison

This section contains the principal realized refinement. Its abstract result is independent of the source construction; its specialization assumes exactly the source orbital and separation bounds stated above. The proof uses physical operators before coefficient expansion and does not take the norm of an unbounded Coulomb multiplier on all of L^2.

Lemma 5.1 (Hardy bound for tensor synthesis)**.** *Let $X : \mathbb{C}^m \to H^1(\mathbb{R}^3)$ and $Y : \mathbb{C}^m \to H^1(\mathbb{R}^3)$ be bounded maps, also viewed as maps into L^2. If V denotes multiplication by $|x-y|^{-1}$, then on the finite coefficient tensor space*

$$\|V(X \otimes Y)\| \le 2\,\|\nabla X\|\,\|Y\|\,.$$

The norms on the right are $\mathbb{C}^m \to L^2$ induced norms; ∇X takes values in vector-valued L^2.

Proof. For every $f \in H^1(\mathbb{R}^6)$, Hardy in x with y fixed, followed by integration in y, gives $\|Vf\| \le 2\,\|\nabla_x f\|$. The same holds componentwise for spin. Products of H^1 functions in independent variables have the required partial weak derivatives. Finite sums follow by linearity. The Hilbert tensor norm identity gives $\|\nabla_x(X \otimes Y)\| = \|\nabla X\|\,\|Y\|$, proving the result, including entangled coefficient vectors rather than only simple tensors. □

Theorem 5.2 (Two-particle Lipschitz estimate in physical maps)**.** *Let $J, Y : \mathbb{C}^m \to H^1(\mathbb{R}^3)$ be exact L^2 isometries. Set*

$$E = J - Y, \quad d_0 = \|E\|\,, \quad d_1 = \|\nabla E\|\,, \quad K_Y = \|\nabla Y\|\,.$$

For the two-particle compressions $v_J = (J \otimes J)^* V (J \otimes J)$ and $v_Y = (Y \otimes Y)^* V (Y \otimes Y)$,

$$\|v_J - v_Y\| \le 2d_1 + 6K_Y d_0. \tag{5.1}$$

The same bound holds after tensoring with spin identities and restricting to the antisymmetric two-particle space.

Proof. Write $A_2 = J \otimes J$ and $B_2 = Y \otimes Y$. Both have norm one, and

$$A_2 - B_2 = E \otimes J + Y \otimes E.$$

Thus

$$\|A_2 - B_2\| \le 2d_0, \qquad \|\nabla_x(A_2 - B_2)\| \le d_1 + K_Y d_0.$$

Hardy gives $\|VB_2\| \le 2K_Y$ and $\|V(A_2 - B_2)\| \le 2d_1 + 2K_Y d_0$. Every displayed map takes the finite coefficient space into the appropriate multiplier domain. Therefore

$$A_2^* V A_2 - B_2^* V B_2 = A_2^* V (A_2 - B_2) + (A_2 - B_2)^* V B_2$$

is an identity of bounded finite operators. Its norm is at most $(2d_1 + 2K_Y d_0) + (2d_0)(2K_Y)$, proving (5.1). Spin tensoring preserves these norms. Since the pair operator and tensor maps commute with exchanging the particles, restriction to the antisymmetric subspace is valid and cannot increase the norm. □

Corollary 5.3 (Fixed-particle-number lift)**.** *For every integer $0 \le N \le 2m$, define $J_N = \bigwedge^N (J \otimes I_2)$ and $Y_N = \bigwedge^N (Y \otimes I_2)$, with normalized exterior products. Then*

$$\|J_N^* W_N J_N - Y_N^* W_N Y_N\| \le \binom{N}{2}(2d_1 + 6K_Y d_0), \qquad W_N = \sum_{a<b} |x_a - x_b|^{-1}. \tag{5.2}$$

Proof. Because J and Y are isometries, compression of a pair term on the full N-fold tensor space leaves identity factors on the $N-2$ spectators. The difference is the corresponding lift of $v_J - v_Y$. Sum over the $\binom{N}{2}$ pairs and apply the triangle inequality. The sum preserves antisymmetry, so its restriction has no larger norm. There is no factor equal to the Fock-space dimension and no extra N from comparing J_N and Y_N directly. For $N < 2$, both interactions and the bound are zero. □

5.1 Normalized disjoint cutoffs and source specialization

Fix a smooth radial ϑ equal to one for $r \le 1/5$ and zero for $r \ge 1/4$, with $0 \le \vartheta \le 1$ and bounded derivative. Let

$$\zeta_i(x) = \vartheta(|x - X_i|/D)\phi_i(x), \qquad Y_i = \zeta_i / \|\zeta_i\|, \qquad Y e_i = Y_i.$$

For large D, radial integration of e^{-2r} and the cutoff derivative give a uniform number

$$\|Y_i - \phi_i\|_{H^1} \le \sigma_D, \qquad \sigma_D = C(1+D)^2 e^{-D/5} \le C e^{-.18D}. \tag{5.3}$$

To justify normalization explicitly, first $\|\zeta_i - \phi_i\|_{H^1} \le C(1+D)^2 e^{-D/5}$ and $|\,\|\zeta_i\| - 1| \le \|\zeta_i - \phi_i\|_2$. For large D the denominator is at least $1/2$. Writing $Y_i - \phi_i = (\zeta_i - \phi_i)/\|\zeta_i\| + (1/\|\zeta_i\| - 1)\phi_i$ proves (5.3) after adjusting C. Since $\|\phi_i\|_2 = 1$ and $\|\nabla\phi_i\|_2 = 1$, the gradient norms of Y_i are uniformly bounded.

The balls of radius $D/4$ are disjoint when $|X_i - X_j| \ge D - 2S$ and $S \le D/100$. Therefore both function supports and gradient supports are disjoint up to null sets. Consequently

$$Y^*Y = I, \qquad K_Y = \|\nabla Y\| = \max_i \|\nabla Y_i\| \le 2 \tag{5.4}$$

after enlarging a fixed lower bound on D. The second equality, which fails for arbitrary overlapping columns, is the key uniform kinetic bound.

Take $J = \mathcal{J}_0$, already an exact isometry. If $a_D = \max_i \|\omega_i - \phi_i\|_{H^1}$, Cauchy–Schwarz in the column sum gives

$$d_0, d_1 \le \|J - Y\|_{\mathbb{C}^m \to H^1} \le \sqrt{m}(a_D + \sigma_D) =: \epsilon_D. \tag{5.5}$$

Indeed $\|\sum_i c_i(\omega_i - Y_i)\|_{H^1} \le \|c\|_2 \left(\sum_i \|\omega_i - Y_i\|_{H^1}^2\right)^{1/2}$. With $a_D \le Cm/D$, this is $\epsilon_D \le Cm^{3/2}/D + C\sqrt{m}e^{-.18D}$. Also $\|\nabla J\| \le K_Y + \epsilon_D = O(1)$ in the source regime, though Theorem 5.2 only needs K_Y and the two difference norms.

Theorem 5.4 (Improved Coulomb compression)**.** *Suppose $m \ge 1$, the separation and orbital assumptions above hold, and D is sufficiently large in the source regime $m + S \le D^{1/4}$. For $N = m$, let $W_{\text{eff}} = \mathcal{J}_m^* W \mathcal{J}_m$ in the abstract spin-orbital space. Then*

$$\left\| W_{\text{eff}} - \tfrac{5}{8}\sum_i n_{i\uparrow} n_{i\downarrow} \right\| \le 14\binom{m}{2}\epsilon_D + \frac{\binom{m}{2}}{D/2 - 2S} + C\lfloor m/2 \rfloor \sigma_D \tag{5.6}$$

$$\le C\left(\frac{m^{7/2}}{D} + \frac{m^2}{D} + m^{5/2}e^{-.18D}\right) \le \frac{Cm^{7/2}}{D}. \tag{5.7}$$

The first line extends to any $0 \le N \le 2m$ with m replaced by N in the binomial and occupancy factors, while ϵ_D still uses the m spatial columns. No single-occupancy assumption is made.

Proof. Equations (5.4)–(5.5) and Corollary 5.3 bound replacement of J by Y by $14\binom{N}{2}\epsilon_D$. Disjoint supports force a nonzero four-function coefficient to have the same site label in the bra and ket at each spatial coordinate. The exact compressed interaction in Y is therefore

$$W_Y = \sum_i U_i n_{i\uparrow} n_{i\downarrow} + \sum_{i<j} D_{ij} n_i n_j, \quad n_i = n_{i\uparrow} + n_{i\downarrow}.$$

Here $U_i = \iint |Y_i(x)|^2 |Y_i(y)|^2 / |x-y| \, \mathrm{d}x \, \mathrm{d}y$ and D_{ij} is the same density integral with Y_i, Y_j. For different sites, support separation yields $0 \le D_{ij} \le (D/2 - 2S)^{-1}$. In a fixed-$N$ occupation vector,

$$\sum_{i<j} n_i n_j = \tfrac{1}{2}\left(N^2 - \sum_i n_i^2\right) \le \binom{N}{2},$$

since each $n_i \in \{0,1,2\}$ satisfies $n_i^2 \ge n_i$. Thus the nonnegative direct operator has norm bounded by the middle term of (5.6); it has not been dropped without a bound.

For completeness, for four H^1 functions Hardy and Cauchy–Schwarz imply

$$\left| \iint \frac{\overline{a(x)b(y)}c(x)d(y)}{|x-y|} \, \mathrm{d}x \, \mathrm{d}y \right| \le 2 \|a\|_2 \|b\|_2 \|\nabla c\|_2 \|d\|_2 .$$

Telescoping the four arguments between Y_i and ϕ_i, using symmetry of the Coulomb form when necessary to place the differentiated difference on either side, gives $|U_i - U| \le C\sigma_D$. The normalized hydrogen density has radial law $4r^2 e^{-2r} \, \mathrm{d}r$, so Newton's radial average gives

$$U = 8 \int_0^\infty r e^{-2r} [1 - e^{-2r}(1 + 2r + 2r^2)] \, \mathrm{d}r = 8(1/4 - 1/16 - 1/16 - 3/64) = 5/8.$$

Pauli exclusion removes equal-spin onsite terms, and the two opposite-spin terms with the usual $1/2$ pair convention produce exactly $U_i n_{i\uparrow} n_{i\downarrow}$. At most $\lfloor N/2 \rfloor$ sites can be doubly occupied, so the onsite error has the stated norm. Adding the three contributions proves the first line. Insert the bound on ϵ_D and absorb exponentials for sufficiently large D to obtain (5.7). When $N = 1$, all three terms vanish exactly. □

5.2 What the refinement changes

The source loses m^4 by summing coefficients; the proof above loses $\sqrt{m}$ in forming the physical H^1 synthesis error and $\binom{m}{2}$ in lifting an actual pair operator. This reduces an algebraic power, not merely an unspecified numerical constant. For a compression-only tolerance ϵ, the displayed sufficient algebraic requirement changes from $D \gtrsim m^5/\epsilon$ to $D \gtrsim m^{7/2}/\epsilon$. It does not follow that the final reduction uses a smaller polynomial degree: $SD^{-1/4}$ and other calibration conditions may dominate. No bit-complexity exponent or wall-clock speedup is asserted. The argument developed in this report is a local tensor-map and form estimate, not a claim of historical priority.

6 Positive angular quadrature with fewer spatial nodes

The second local refinement concerns the source's `lem:hopping-quadrature` in S3. Its $\mu = \cos\theta$ parametrization has a square-root modulus at the poles and therefore uses $O((D/h)^4)$ radial-angular cells to achieve diameter $O(h)$. Uniform polar angles remove that particular loss. The tradeoff is that angular weights are no longer exact rationals. The following proof accounts for their precision rather than treating them as exact.

Theorem 6.1 (Positive quadrature node bound)**.** *Let $D \geq 1$ be an integer and $0 < \epsilon < 1/10$ a rational tolerance. Let χ be the source's fixed rational piecewise-polynomial cap and let L be a real function on $|v| \leq D/8$ with a fixed known Lipschitz bound L_0. Suppose the source's deterministic sample algorithm approximates $L(v)$ to absolute error ξ with polynomial cost in D, m, ξ^{-1} and the input bit length, including coordinate errors. Assume also $|L(v)| \leq C(D+m)$ and*

$$I = e^D T \geq \kappa_0 > 0, \qquad T = \frac{1}{\pi} \int_{|v| \leq D/8} \frac{1 - \chi(|v|/D)}{|v|} e^{L(v)} \, \mathrm{d}^3 v,$$

with a known positive rational constant lower bound κ_0. Then a positive rational $\widehat{I}$ satisfying

$$\left| \log \frac{\widehat{I}}{I} \right| \leq \epsilon \tag{6.1}$$

can be constructed using $O((D/\epsilon)^3)$ spatial sample nodes for fixed L_0. Its weight arithmetic has polynomial bit cost. For the source reference contacts the hypotheses hold, using its sample-log evaluation and positive reference lower bound. The number of ray samples per spatial node is unchanged.

Proof. Write $v = r\omega(\theta, \varphi)$ with

$$\omega = (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta), \qquad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi \leq 2\pi.$$

The angular derivatives have norms at most one, so $|\omega(\theta, \varphi) - \omega(\theta', \varphi')| \leq |\theta - \theta'| + |\varphi - \varphi'|$ within any parametrization rectangle, including a rectangle next to a pole. Choose a uniform rational radial baseline mesh of spacing at most $h \leq 1$ and comparable to h, then insert the finitely many cap breakpoints. Use equal angular meshes $\theta_j = j\pi/M$, $\varphi_k = 2k\pi/N$, with $M, N = \lceil CD/h \rceil$ for a fixed adequate C. There are $O(D/h)$ radial intervals and $MN = O((D/h)^2)$ angular rectangles. Since $r \leq D/8$, every spatial cell has diameter at most $C_0 h$. The total number is $O((D/h)^3)$. Pole multiplicities affect only null sets, and no Jacobian singularity is used to control cell diameter.

For a radial interval I_a the exact weight

$$w_a = \int_{I_a} (1 - \chi(r/D)) r \, \mathrm{d}r \geq 0$$

is rational and computed by polynomial integration. Zero weights are omitted. For one angular rectangle, including the factor $1/\pi$, the exact weight is

$$\begin{aligned} a_j &= \frac{2}{N} \big(\cos(j\pi/M) - \cos((j+1)\pi/M) \big) \\ &= \frac{4}{N} \sin\!\left(\frac{(2j+1)\pi}{2M} \right) \sin\!\left(\frac{\pi}{2M} \right) \geq \frac{4}{NM^2} > 0. \end{aligned} \tag{6.2}$$

The last inequality uses $\sin(\pi/(2M)) \geq 1/M$, which follows from concavity of sine on $[0, \pi/2]$, and the fact that the other sine is at least the endpoint sine. This includes $M = 1$. Telescoping gives $\sum_{j,k} a_j = 4$.

Choose one sample v_{ajk} per cell. For the positive exact-weight sum $Q = \sum_{a,j,k} w_a a_j e^{L(v_{ajk})}$, the log-Lipschitz bound yields

$$e^{-L_0 C_0 h} Q \leq T \leq e^{L_0 C_0 h} Q. \tag{6.3}$$

No logarithm of $1 - \chi$ or of the radial weight is taken.

The a_j are generally irrational. Fix $0 < \eta < 1/2$ and compute positive rational approximations with

$$|\widehat{a}_j - a_j| \le \eta \frac{4}{NM^2}.$$

Then $(1-\eta)a_j \le \widehat{a}_j \le (1+\eta)a_j$. Absolute sine/cosine precision $O(\log N + \log M + \log(1/\eta))$ bits, with fixed constant factors, suffices on bounded angular intervals. The sine-product form avoids a needless subtraction of nearly equal cosines; a certified absolute-error cosine method would also be valid. Rational enclosures for π and elementary functions provide a deterministic finite arithmetic procedure.

If the approximate sample log $\widehat{L}_{ajk}$ satisfies $|\widehat{L}_{ajk} - L(v_{ajk})| \le \xi$, positivity implies

$$(1-\eta)e^{-\xi}e^{D}Q \le \widetilde{Q} := \sum_{a,j,k} w_a \widehat{a}_j e^{D+\widehat{L}_{ajk}} \le (1+\eta)e^{\xi}e^{D}Q.$$

Combining with (6.3) bounds the logarithmic error by

$$L_0 C_0 h + \xi - \log(1-\eta). \tag{6.4}$$

The relative weight loss is independent of the exponentially small unscaled integral. The sum of approximate non-exponential weights obeys

$$W_0 := \sum_{a,j,k} w_a \widehat{a}_j \le (1+\eta)\, 4 \int_0^{D/8} r \, \mathrm{d}r \le CD^2.$$

Approximate each scaled exponential to absolute error at most $\kappa_0\epsilon/(100\max(1, W_0))$ and clip any negative approximation to zero. The sum error is at most $\kappa_0\epsilon/100$; clipping does not increase the error to a positive true value. The exponent arguments have size $O(D+m)$, so the working bit length is polynomial in $D, m, \log(1/\epsilon)$ and the input size. The resulting rational sum is positive because $I \ge \kappa_0$ and the preceding errors are smaller than a fixed fraction of that lower bound.

Choose, for example, $h \le \epsilon/[20(1+L_0C_0)]$, $\xi \le \epsilon/20$, and $\eta = \epsilon/20$. Since $-\log(1-\eta) \le 2\eta$, (6.4) is at most $\epsilon/5$. The last additive rounding error contributes less than $\epsilon/10$ to the log error for the stated range, leaving margin in (6.1). All choices have polynomial cost and give $O((D/\epsilon)^3)$ spatial cells. □

For M557, the source proves $e^{D}T_{ij}(Dp) \ge 2e^{-2}$ for sufficiently large D, so a fixed rational lower bound such as $\kappa_0 = 1/8$ is valid. Its capped-field Lipschitz sample algorithm, with $O(m/\epsilon)$ ray panels, supplies the log approximations. Thus the improvement is the spatial-node exponent three instead of four at the same inverse-polynomial relative tolerance. All field evaluations, ray panels, arithmetic, and global reduction steps still have their own costs. No optimal cubature theorem or measured total-runtime improvement is asserted. This is classical positive, measure-adapted quadrature in the source's positive integral representation, not a new Ram-specific integration law.

7 Complement exclusion without a ground eigenvector

Theorem 7.1 (Sharp scalar block bound)**.** *Let T be a lower-bounded self-adjoint operator, $P \ne 0$ a finite-rank orthogonal projection with* $\operatorname{Ran} P \subset D(T)$*, and $Q = I - P$. Let P, Q preserve the form domain. Suppose*

$$PTP|_{\operatorname{Ran} P} \succeq eI, \qquad QTQ \succeq cQ, \qquad \|QTP\| \le \beta,$$

where $e, c \in \mathbb{R}$ and $\beta \geq 0$. Then

$$\inf \operatorname{spec} T \geq L(e, c, \beta) := \frac{e + c - \sqrt{(c-e)^2 + 4\beta^2}}{2}. \tag{7.1}$$

If $E_P = \min \operatorname{spec}(PTP|_{\operatorname{Ran} P}) < c$, then

$$0 \leq E_P - \inf \operatorname{spec} T \leq \ell(E_P, c, \beta) := \frac{\sqrt{(c-E_P)^2 + 4\beta^2} - (c - E_P)}{2} \leq \frac{\beta^2}{c - E_P}. \tag{7.2}$$

These bounds are optimal given only the three scalar block data.

Proof. For a unit form-domain vector $u = p + q$, write $s = \|p\|$, $t = \|q\|$. The domain hypothesis makes the mixed form equal to the operator pairing with Tp, so

$$q_T[u] \geq es^2 + ct^2 - 2\beta st, \qquad s^2 + t^2 = 1.$$

The smaller eigenvalue of the real matrix $\left(\begin{smallmatrix} e & -\beta \\ -\beta & c \end{smallmatrix}\right)$ is (7.1); its lower bound holds for every s, t and therefore every admissible continuum vector. Taking the infimum proves the claim. Restriction to $\operatorname{Ran} P$ gives $\inf \operatorname{spec} T \leq E_P$. Put $d = c - E_P > 0$. The loss obeys

$$\ell = \frac{2\beta^2}{\sqrt{d^2 + 4\beta^2} + d} \leq \frac{\beta^2}{d},$$

with value zero when $\beta = 0$. A two-dimensional matrix with diagonal entries E_P, c and off-diagonal $-\beta$ attains equality in the exact formula, proving optimality of the scalar information. No minimizing continuum vector was used. □

Corollary 7.2 (Interval form of a NO certificate)**.** *Suppose validated finite data give $e_- \leq E_P \leq e_+$, an analytic complement certificate gives c_-, and $\beta \leq b_+$. Then*

$$L(e_-, c_-, b_+) \leq \inf \operatorname{spec} T \leq e_+.$$

If $e_+ < c_-$, the leakage satisfies $E_P - \inf \operatorname{spec} T \leq \ell(e_+, c_-, b_+) \leq b_+^2/(c_- - e_+)$. For a proposed lower threshold $z < c_-$, the two rational inequalities

$$e_- \geq z, \qquad (e_- - z)(c_- - z) \geq b_+^2 \tag{7.3}$$

are sufficient to prove $\inf \operatorname{spec} T \geq z$ without evaluating a square root.

Proof. The smaller eigenvalue is monotone in each diagonal entry under the matrix order and decreases as the magnitude of the off-diagonal increases; the latter follows directly from the formula. The loss increases as E_P increases, as c decreases, or as β increases. These facts prove the interval bounds. Finally (7.3) and $c_- - z > 0$ are exactly the two-by-two positive-semidefinite Schur condition for the comparison matrix minus zI. □

7.1 A direction-sensitive physical-Gram certificate

Proposition 7.3 (Residual-matrix lower certificate)**.** *Use the notation of Proposition 4.1 for a semibounded T, and suppose $QTQ \succeq cQ$ as a form, with $Q = I - P$ preserving its domain. For any real $z < c$, if*

$$K - zG - \frac{F^*F}{c - z} \succeq 0, \tag{7.4}$$

then $T \succeq zI$ on its whole form domain. This finite certificate is coordinate invariant. It does not require QTQ to have discrete spectrum.

Proof. Write $u = Sa+v$ with $v \in \operatorname{Ran} Q$ in the form domain. Orthogonality and the operator-domain condition on Sa give

$$\begin{aligned} q_{T-z}[u] &\geq a^*(K - zG)a + (c - z)\|v\|^2 + 2\Re\langle v, Fa\rangle \\ &= (c - z)\left\|v + (c - z)^{-1}Fa\right\|^2 + a^*\left(K - zG - \frac{F^*F}{c - z}\right)a \geq 0. \end{aligned}$$

The square is an identity in the Hilbert norm and does not assert that Fa is a form-domain vector. It only bounds the already well-defined form. Gram congruence proves invariance. □

This can improve a scalar certificate when the largest residual acts only on high finite-energy directions. Replacing F^*F by $\beta^2 G$ recovers the scalar condition. It remains a sufficient test, not the exact Feshbach map unless the actual complementary resolvent is retained. Classical Feshbach–Schur theory is a predecessor [DSS]; the local contribution is the explicit physical-metric and residual account at this source interface.

As an exact finite diagnostic, take retained block $\operatorname{diag}(0, 10)$, complementary scalar $c = 2$, and coupling row $(0, 1)$. At $z = 0$, (7.4) is $\operatorname{diag}(0, 10) - \frac{1}{2}\operatorname{diag}(0, 1) \succeq 0$, proving a zero lower bound. The scalar data $e = 0, c = 2, \beta = 1$ only give $1 - \sqrt{2} < 0$. The full three-by-three example has the uncoupled zero eigenvector and a positive remaining two-by-two block. This demonstrates a certificate advantage, not a Coulomb computation.

7.2 Source complement bound in the correct energy units

Apply Theorem 7.1 with $T = \mathcal{H}$ and its finite-rank projection equal to Π. Retain the distinct one-particle projection $P_1 = P_0 \otimes I_2$, and let

$$\bar{\beta} = m\varepsilon + C_H m^2/Z_*, \quad c_- = g - 2m\varepsilon, \quad \varepsilon = \|AP_1\| \leq C_R\sqrt{m}e^{-.75D}.$$

Here C_H, C_R are valid bounds for the source estimates, not fitted constants. The source also gives $\|\Pi\mathcal{H}\Pi\| \leq \bar{\beta}$, so $E_\Pi \leq \bar{\beta}$. For sufficiently large D, $\bar{\beta} < c_-$. Theorem 7.1 gives

$$0 \leq E_\Pi - \inf\operatorname{spec}\mathcal{H} \leq \ell(E_\Pi, c_-, \bar{\beta}) \leq \frac{\bar{\beta}^2}{c_- - \bar{\beta}}. \tag{7.5}$$

For comparison to the source's fixed $c = g/2$ square completion, retain that same c and use $E_\Pi < c$. When $E_\Pi \leq c/2$, the new rational bound is at most $2\bar{\beta}^2/c$, with a strict improvement in the exact formula except degenerate cases. As $E_\Pi \to 0$ and $\bar{\beta} \to 0$, its leading loss is $\bar{\beta}^2/c$ instead of the source's $2\bar{\beta}^2/c$. This is a classical optimization of the scalar bound, not by itself the principal contribution rating.

Multiplying a valid loss by Z_* gives, with fixed denominators bounded below,

$$Z_*\frac{\bar{\beta}^2}{c_- - \bar{\beta}} \leq C\left(m^3e^{-.5D} + m^{7/2}e^{-.75D} + m^4e^{-D}\right). \tag{7.6}$$

The full one-body residual cannot be multiplied linearly by Z_*: $e^{-.75D}e^D$ grows. Its squared use against the complement gap is essential. Existence of E_Π is finite dimensional; no full-space ground eigenvector or uniform many-body excitation gap is assumed.

8 Revised simulation statement and remaining dependencies

Corollary 8.1 (Local replacement in the source error budget)**.** *Assume the source one-body matrix estimate* (2.6)*, physical orbital/separation estimates, one-particle complement gap, and fermionic Hardy estimate. With* $\bar{\beta}, c_-$ *as above and* $\bar{\beta} < c_-$*,*

$$\begin{aligned} |E_0/Z_* + mZ_*/2 - E_{\mathrm{Hub}}(t)| &\le m\delta_1 + \frac{Cm^{7/2}}{D} + Z_*\ell(E_\Pi, c_-, \bar{\beta}) \\ &\le m\delta_1 + \frac{Cm^{7/2}}{D} + \frac{Z_*\bar{\beta}^2}{c_- - \bar{\beta}}. \end{aligned} \tag{8.1}$$

The explicit first line of (5.6) *may replace* $Cm^{7/2}/D$*. The conclusion concerns the full continuum spectral infimum.*

Proof. The exact finite compression of $Z_*\mathcal{H}$ is $d\Gamma(Z_*\mathcal{J}^*A\mathcal{J}) + W_{\mathrm{eff}}$. Its difference from the half-filled Hubbard operator has norm at most $m\delta_1 + Cm^{7/2}/D$, by the elementary m-particle one-body lift and Theorem 5.4. Finite variational minima differ by no more than that norm. Removing the finite restriction costs at most the loss from (7.5), multiplied by Z_*. Equation (1.2) then proves the statement. □

If the right side is at most $\alpha\gamma/64$, the source threshold transfer in Section 2 remains valid with its existing margins. This is a conditional improvement at an important local analytic step. It is not an independent reduction from QMA, a new proof of the one-well eigenvalue oracle, or a proof that the compressed Hubbard minimum can be computed in polynomial time. Even an exact coordinate representation leaves the many-electron dimension $\binom{2m}{m}$ and the hard minimization intact.

The remaining dependencies are concrete. The imported square-lattice hardness theorem must supply the claimed polynomial interaction scales. The source's analytic constants must be uniform through moving poles, cap transitions, rational centers, and the tuning cube. The relative tail and quadrature estimates must survive Z_* amplification in their specified regions. The complement gap must hold for all form-domain vectors, not just sampled functions. The Ritz oracle needs certified arithmetic and an actual mass lower bound. These are not supplied by a Ram change of representation; they are the source's separately argued obligations. No defect in these obligations is asserted merely because they were not independently re-proved here.

9 Comparisons, benefits, and limits

9.1 Primary public literature and the 2026 boundary

The public literature was checked by public mathematical titles and identifiers only, on October 8, 2026 UTC. No private text was used in a public query. O'Gorman–Irani–Whitfield–Fefferman's published Theorem 1 is a fixed-basis result; Appendix D, Corollary 1 permits positive unit point charges while retaining that fixed-basis setting [OIW F]. Their Section V discusses the complete-space issue. Thus that predecessor does not alone supply M557's omitted-state lower bound.

Cubitt–Montanaro–Piddock identify field-free Heisenberg hardness on the square lattice via the spatially sparse branch [CMP]. Schuch–Verstraete's earlier continuum/Hubbard construction uses an external magnetic-field setting [SV]; it should not be conflated with the nuclear-only input. Feshbach–Schur elimination and quantitative perturbation bounds are established predecessors [DSS]. Our scalar theorem is proved directly for the spectral infimum instead of importing an isolated-eigenvalue hypothesis.

Schleich and collaborators' 2026 work addresses chemically motivated simulation through physically prepared states and dynamics, with an explicitly different problem formulation [Sch+]. Its accessible-process perspective does not show that every promised adversarial Coulomb ground-energy instance is easy. This report makes neither a hardness-versus-chemistry contradiction nor a comprehensive historical-novelty claim. The literature search establishes these comparisons, not absence of all other relevant literature.

9.2 The actual M558 comparison

The companion M558 is *QMA-hardness of continuum Coulomb energy with unit nuclear charges*. Only its introduction and main theorem were read for this comparison; its full proof is a separate assignment. Its stated inputs have all $Z_\alpha = 1$, polynomially many nuclei and electrons, rational positions, all spin sectors, and no supplied orbital basis. Its introduction describes a positive continuous-charge density, polynomial confinement, and discretization to unit nuclei. It starts from the external spin source rather than invoking M557's hardness theorem.

As a logical statement, if M558's theorem is valid, hardness for the unit-charge subclass implies hardness for the larger binary-charge input class. Conversely, replacing each M557 charge by that many unit nuclei can require exponentially many nuclei and is not a polynomial output conversion. M557's hydrogenic exponential-accuracy and relative-tunneling construction is quantitatively distinct. Our $m^{7/2}/D$ refinement is proved for M557's close-to-hydrogen modes and widely separated cutoffs; it is not silently transferred to M558's different local-mode or occupancy construction.

9.3 Advantages and costs of the local approach

The map-level estimate is short enough to isolate every norm conversion, does not sum an m^4 coefficient array, preserves fermionic/spin quantifiers, and gives an explicit general-N extension. Normalization of the cutoff modes is essential: the spectator factors in Corollary 5.3 are identities only for exact isometries. A proof using raw nonorthogonal hydrogen modes in that step would need additional Gram terms.

The scalar complement bound is optimal for its three numbers and easy to validate by rational inequalities. The direction-sensitive matrix certificate can be materially better when residuals avoid the lowest finite-energy directions. Its cost is that the matrices may still be exponentially large on m particles, and continuum residual enclosures can be difficult. Neither certificate is a polynomial-time algorithm for the QMA-hard finite minimum.

The estimates are sufficient and deliberately conservative. The $\sqrt{m}$ column conversion may not be sharp; better near-orthogonality of derivative errors could improve it, but no such bound is assumed. Positivity of direct density terms could be used one-sidedly for lower estimates; here we bound them in norm to preserve a two-sided simulation comparison. A smaller displayed exponent does not automatically reduce the source's dominating calibration scale.

10 Verification, assessment, and future work

10.1 Executed checks and what they establish

The bounded author check suite is local and deterministic. It uses exact rational arithmetic for Gram projection, residual decomposition, scalar-shift diagnostics, onsite 5/8, and the two-site superexchange identity. Bounded finite Hermitian examples exercise the scalar block bound and the direction-sensitive certificate; a finite matrix analogue tests the tensor-difference identity and the

pair-lift combinatorial bound. Angular weight normalization and lower bounds are checked at finitely many rational-cosine meshes. Those tests are algebraic diagnostics, not numerical evidence for Hardy's continuum inequality or the source's large-D constants. The analytic results in Sections 5–8 rest on the displayed proofs.

The author suite `_audit/M557-author-checks.py` executed 114 exact rational assertions with Python 3.12.14; all passed. Separate corroborating finite checks and targeted internal mathematical review are retained in the local audit records with their precise read and execution scopes. They are not external peer review, a full replay of the reduction, or verification of the entire source theorem.

The source and private-input hashes, exact read scope, executed commands, check results, compiler responses, build diagnostics, visual QA, internal review, and artifact hashes are recorded in the assigned audit/build paths. The full manuscript source is standalone: it has no external mathematical input files or required image assets. No package or TeX installation was performed.

10.2 Evidence-based contribution assessment

The realized contribution is rated C3/5: the physical two-particle argument removes a dimension-dependent coefficient-counting loss, proves the $m^{7/2}/D$ local bound and a general particle-number form, and integrates it into the continuum error budget. The positive angular quadrature improves the sufficient spatial-node exponent from four to three with an explicit rational-weight budget. The sharp scalar and matrix-residual certificates clarify another local step and its input requirements. Together these are notable local refinements, not merely relabeling or a numerical factor.

A C4 or C5 conclusion is not claimed. The report does not establish a global improved scale, a faster reduction, a newly validated Coulomb oracle, a new hardness theorem, or historical novelty. Applicability was screened separately in the catalogue; its L5 rating is not used as contribution evidence. The evidence labels for the internal contribution record are `thm:pair`, `thm:compression`, `thm:quadrature`, `thm:block`, `prop:matrix-NO`, and `cor:simulation`.

10.3 Concrete future investigations

First, a validated enclosure implementation could bound G, K, and the full residual Gram matrix for a small actual tuned instance, using interval integration and separate complement estimates. It would need to retain uncertainty from the orbital approximation and form domain; checking only finite eigenvalues would be insufficient. Second, derivative-overlap estimates might replace the columnwise $\sqrt{m}$ loss by an operator estimate exploiting separation. Third, one could propagate the refined compression term through every calibration constraint to see whether a lower total polynomial scale is genuinely possible. Fourth, matrix-aware residual certificates could be compared with scalar ones on finite examples using identical certified tolerances. None of these computations or improvements has been completed here.

A Input and provenance ledger

All paths in this appendix are local. The report root is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/
```

The assigned output is `M557-Coulomb-Binary/TR-M557-Ram-Coulomb-Binary.tex` and the matching PDF. The contribution policy is `contribution-policy.json`; the input pin comes from row 557 of `00-Applicability-Catalogue/applicability.json`. The audit records are under `_audit/M557-author*`, with the single-record score file `_audit/contribution-assessment-557.json`.

Rendering and build logs are under `_build/TR-M557-Ram-Coulomb-Binary/`. No shared catalogue, queue, helper, progress, original source, or other report is edited by this author.

Input	Actual read scope and use
M557 main and preamble	Entire files; all included section files identified; no uninspected mathematical appendices.
M557 S0	All 133 lines; input, theorem, scope, comparisons, mechanism.
M557 S1	All 697 lines; finite perturbation, signed mediators, geometry/right inverse, Hubbard proof.
M557 S2	All 958 lines; well/domain/tail proofs, analytic patches, graded Ritz, exact conditioning and arithmetic, rational tuning.
M557 S3	All 780 lines; relative barriers, surrogate replacement, positive quadrature, displacement/orientation, calibration.
M557 S4	All 767 lines; Gram and full residual, IMS complement, many-body Hardy/elimination, compression, simulation.
M557 S5	All 161 lines; fixed global scale, computation order, output bits, rational thresholds.
M557 references/README	Entire files; citation and provenance checks.
M558 introduction	Lines 1–150 only; main theorem, input conventions, companion distinction and beginning of strategy; no proof verification.
Volume 21, ch01a	Entire file; domain and bounded coordinate equivalence.
Volume 21, ch01d	Entire file; physical metric, full residual, Gram equivalence and finite-information limits.
Volume 21, ch01f	Entire file; pollution example, strong/norm distinction, positive form statement and domain-aware inversion.
TR-core-pat-5	Lines 310–466; polynomial evaluation, coarse lifts, Schur residual, Woodbury and conditioning limits.

SHA-256 values are stored in the machine-readable author input ledger. Hashes identify bytes read, not the truth of claims. The artifact manifest records the report source, PDF, executed check files, internal-review evidence, and input hashes. No claim that the full confidential corpus was read is made.

References

[M557] OpenAI. *Continuum Coulomb hardness with binary nuclear charges.* September 24, 2026. Local pinned source, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Source path and read scope in Section 1 and Appendix A.

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